



INTERNATIONAL UNIVERSITY OF BUSINESS AGRICULTURE AND TECHNOLOGY

Department of Computer Science and Engineering

ASSIGNMENT TITLE

Emergency Potable Water Allocation for a Flood-Affected Municipality

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Programme

Bachelor of Computer Science and Engineering (BCSE)

Course Code and Title

Structured Programming LAB

Section

D

Course Instructor

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2. Problem Analysis

After a flood, the municipality has only a limited amount of potable water to distribute among six zones (a hospital, two emergency shelters, two residential wards, one fire/emergency service), each with different requested quantity, minimum survival need, distribution loss %, and waiting cycles. Total demand exceeds supply, so the system must decide **which zones to serve first and how much to give them**. Key conflicts include functional urgency (hospital/fire) vs. numeric need (residential gap), distribution loss (more water must be released than delivered), and fairness (zones waiting longer should not be starved indefinitely).

3. Proposed Solution

A weighted priority score is used via the function `calculatePriority()`:

$$\text{Priority} = \text{Urgency} + (\text{Waiting} \times 2.0) + (\text{Requested} - \text{Minimum})/500 - (\text{Loss} \times 0.10)$$

- **Urgency (fixed):** Hospital/Emergency Service = 10, Shelter = 8, Residential = 5 — keeps life-critical zones ranked highest.
- **Waiting score:** Rewards zones skipped in past cycles, preventing starvation.
- **Demand score:** Reflects how large the unmet gap is.
- **Loss penalty:** Slightly discourages zones that waste more water in transit.

Zones are sorted highest-to-lowest priority (ties broken by more waiting cycles, then smaller minimum). Water is then released in that order: try the **full** request first, else the **minimum**, else whatever water is left — giving partial relief instead of an all-or-nothing outcome.

4. Programme Design

Data structure: One struct `Zone` array (size `TOTAL_ZONES = 6`) holding both input fields (`id`, `name`, `type`, `requested`, `minimum`, `loss`, `waiting`) and computed results (`priority`, `order`, `allocated`, `delivered`, `shortage`, `status`).

Key functions: `loadOriginalData()`, `resetResults()`, `calculatePriority()`, `sortZones()`, `allocateWater()`, `calculateResults()`, `runAllocation()`, `displayResult()`, `searchZone()`, `findHighestShortage()`, `case1()`–`case7()` as scenario drivers called from a menu-driven `main()`.

Flow: Load data → copy for the chosen case → reset → calculate priority → sort → allocate water → compute loss/delivered/shortage/status → display table & summary → return to menu.

5. Optimisation Considerations

- **Single Pass Calculations:** Priority and result calculations each run in a single loop pass per function (no repeated traversals).

- **Local Case Copying:** Only one master copy of original data is kept in main(); each case copies it locally, avoiding permanent duplication or reloading.
- **State Reset Efficiency:** resetResults() clears fields instead of reloading the whole dataset.
- **Early Loop Termination:** Search/match loops break immediately on finding a result.
- **Compact Sort Logic:** Bubble sort is acceptable given the small, fixed size (6 records); tie-breaking is handled in the same comparison pass.

6. Implementation and Results

Techniques used: Structures, arrays of structures, string handling (strcpy/strcmp), modular functions, bubble sort with tie-breaking, menu-driven switch/do-while, formatted printf tables.

Case 1 output (13,500 L available) — Priority Order: Z04 (13.80) → Z03 (13.20) → Z01 (11.80) → Z02 (11.60) → Z05 (11.50) → Z06 (10.90)

ID	Requested	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z04	3800	1800	13.80	4318.18	3800.00	0.00	FULL
Z03	4500	2000	13.20	4891.30	4500.00	0.00	FULL
Z01	4000	3000	11.80	4081.63	4000.00	0.00	FULL
Z02	3500	2500	11.60	208.89	200.53	3299.47	BELOW MIN
Z05	2600	1600	11.50	0.00	0.00	2600.00	UNSERVED
Z06	2000	1500	10.90	0.00	0.00	2000.00	UNSERVED

Summary: Requested 20,400 L | Allocated 13,500 L | Delivered 12,500.53 L | Unresolved 7,899.47 L
| 3 FULL, 0 MIN, 1 BELOW MIN, 2 UNSERVED.

7. Testing and Validation

Case	Purpose	Result
1	Original data, 13,500 L	3 FULL, 1 BELOW MIN, 2 UNSERVED (table above)
2	Water = full requirement of all zones + buffer	All 6 zones reach FULL
3	Water below combined minimum	Only top-priority zones served; rest UNSERVED
4	Z03 & Z04 given equal data	Equal priority (14.00 each); tie-break rule applied
5	Increase loss % of a residential zone	Priority drops slightly; more release water needed

Case	Purpose	Result
6	Increase waiting cycles of an underserved zone	Priority rises; zone may move up and get served
7	User-chosen water quantity	Consistent, reconciled totals for any input
Search (8)	Look up zone by ID	Correct record shown, or "not found" for invalid ID
Highest Shortage (9)	Zone with largest unmet need	Z05 (2600 L shortage) — highest after Case 1

Note: Invalid inputs (zero/negative water, invalid loss %, non-residential zone in Case 5) are correctly rejected by input validation logic.

Screenshot of output-

The screenshot displays a C code editor on the left and its output window on the right. The code defines a structure for water zones and includes function prototypes for loading data and resetting results. The output window shows the execution of 'CASE 1' with an original dataset of 13,500 L available water. It presents an 'ALLOCATION RESULT' table with columns for ID, Zone, Request, Minimum, Priority, Allocated, Delivered, Shortage, and Status. The table lists six zones: Ward 5, Ward 3, Municipal Hospital, Central Flood Shelter, School Emergency Shelter, and Fire and Emergency Service. A 'Priority Order' list follows, ranking the zones by priority. A 'Summary' section provides totals for requested, allocated, and delivered water, as well as unresolved requirements and remaining water. Finally, 'Service Conditions' are listed, showing that full requirements are satisfied for 2 zones, minimum requirements for 1, and completely unserved for 2 zones. The output concludes with the title 'EMERGENCY POTABLE WATER ALLOCATION'.

```

1  #include <stdio.h>
2  #include <string.h>
3  #define TOTAL_ZONES 6
4  struct Zone
5  {
6      char id[10];
7      char name[40];
8      char type[25];
9      float requested;
10     float minimum;
11     float loss;
12     int waiting;
13     float priority;
14     int order;
15     float allocated;
16     float lossAmount;
17     float delivered;
18     float shortage;
19     char status[20];
20 };
21 /* Function Prototypes */
22 void loadOriginalData(struct Zone zones[TOTAL_ZONES]);
23 void resetResults(struct Zone zones[TOTAL_ZONES]);

```

```

===== CASE 1 =====
Original Dataset
Available Water: 13,500 L

===== ALLOCATION RESULT =====
ID   Zone              Request  Minimum  Priority  Allocated  Delivered  Shortage  Status
-----
Z04  Ward 5              3800     1800     13.80    4318       3800       0         MINIMUM
Z03  Ward 3              4500     2000     13.20    4891       4500       0         FULL
Z01  Municipal Hospital  4000     3000     11.80    4082       4000       0         FULL
Z02  Central Flood Shelter 3500     2500     11.60    209        201        3299     BELOW MIN
Z05  School Emergency Shelter 2600     1600     11.50    0          0          2600     UNSERVED
Z06  Fire and Emergency Service 2000     1500     10.90    0          0          2000     UNSERVED

Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
-----
Total Requested Water      : 20400.00 L
Total Allocated Water      : 13500.00 L
Total Effective Water Delivered : 12500.53 L
Total Unresolved Requirement : 7899.47 L
Remaining Water            : -0.00 L

Service Conditions
-----
Full Requirement Satisfied : 2 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum              : 1 zone(s)
Completely Unserved        : 2 zone(s)

===== EMERGENCY POTABLE WATER ALLOCATION =====

```

Lab_assignment.c - Code::Blocks 25.03
File Edit View Search Project Build Debug

```

1 #include <stdio.h>
2 #include <string.h>
3 #define TOTAL_ZONES 6
4 struct Zone
5 {
6     char id[10];
7     char name[40];
8     char type[25];
9     float requested;
10    float minimum;
11    float loss;
12    int waiting;
13    float priority;
14    int order;
15    float allocated;
16    float lossAmount;
17    float delivered;
18    float shortage;
19    char status[20];
20 };
21 /* Function Prototypes */
22 void loadOriginalData(struct Zone);
23 void resetResults(struct Zone);

```

D:\CodeBlocks\Lab_assignment.exe
10. Exit
Enter your choice: 2
===== CASE 2 =====
Enough Water for Complete Requested Quantity
Available Water: 22194.00 L
=====
ALLOCATION RESULT
=====

ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z04	Ward 5	3800	1800	13.80	4318	3800	0	MINIMUM
Z03	Ward 3	4500	2000	13.20	4891	4500	0	FULL
Z01	Municipal Hospital	4000	3000	11.80	4082	4000	0	FULL
Z02	Central Flood Shelter	3500	2500	11.60	3646	3500	0	FULL
Z05	School Emergency Shelter	2600	1600	11.50	2737	2600	0	FULL
Z06	Fire and Emergency Service	2000	1500	10.90	2020	2000	0	FULL

=====
Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)
=====
Summary
=====
Total Requested Water : 20400.00 L
Total Allocated Water : 21694.00 L
Total Effective Water Delivered : 20400.00 L
Total Unresolved Requirement : 0.00 L
Remaining Water : 500.00 L
=====
Service Conditions
=====
Full Requirement Satisfied : 5 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum : 0 zone(s)
Completely Unserved : 0 zone(s)
=====

D:\CodeBlocks\Lab_assignment.c

Lab_assignment.c - Code::Blocks 25.03
File Edit View Search Project Build Debug

```

1 #include <stdio.h>
2 #include <string.h>
3 #define TOTAL_ZONES 6
4 struct Zone
5 {
6     char id[10];
7     char name[40];
8     char type[25];
9     float requested;
10    float minimum;
11    float loss;
12    int waiting;
13    float priority;
14    int order;
15    float allocated;
16    float lossAmount;
17    float delivered;
18    float shortage;
19    char status[20];
20 };
21 /* Function Prototypes */
22 void loadOriginalData(struct Zone);
23 void resetResults(struct Zone);

```

D:\CodeBlocks\Lab_assignment.exe
9. Find Highest Unresolved Requirement
10. Exit
Enter your choice: 3
===== CASE 3 =====
Enter available water in litres: 8000
=====
ALLOCATION RESULT
=====

ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z04	Ward 5	3800	1800	13.80	4318	3800	0	MINIMUM
Z03	Ward 3	4500	2000	13.20	2174	2000	2500	MINIMUM
Z01	Municipal Hospital	4000	3000	11.80	1508	1478	2522	BELOW MIN
Z02	Central Flood Shelter	3500	2500	11.60	0	0	3500	UNSERVED
Z05	School Emergency Shelter	2600	1600	11.50	0	0	2600	UNSERVED
Z06	Fire and Emergency Service	2000	1500	10.90	0	0	2000	UNSERVED

=====
Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)
=====
Summary
=====
Total Requested Water : 20400.00 L
Total Allocated Water : 8000.00 L
Total Effective Water Delivered : 7277.75 L
Total Unresolved Requirement : 13122.25 L
Remaining Water : 0.00 L
=====
Service Conditions
=====
Full Requirement Satisfied : 0 zone(s)
Minimum Requirement Satisfied: 2 zone(s)
Below Minimum : 1 zone(s)
Completely Unserved : 3 zone(s)
=====

D:\CodeBlocks\Lab_assignment.c

Lab_assignment.c - Code::Blocks 25.03

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1#include <stdio.h>

2#include <string.h>

3#define TOTAL_ZONES 6

4struct Zone

5{

6char id[10];

7char name[40];

8char type[25];

9float requested;

10float minimum;

11float loss;

12int waiting;

13float priority;

14int order;

15float allocated;

16float lossAmount;

17float delivered;

18float shortage;

19char status[20];

20};

21/* Function Prototypes */

22void loadOriginalData(struct Zone

23void resetResults(struct Zone

Logs & others

Code::Blocks X Search results X Cccc X

File

Line

Message

=== Build file: "no

=== Build finished:

D:\CodeBlocks\Lab_assignment.c

D:\CodeBlocks\Lab_assignment.exe

===== CASE 4 =====

Equal Priority Test

Z03 and Z04 have been modified to create equal priority.

=====

ALLOCATION RESULT

=====

ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z03	Ward 3	4000	2000	14.00	4444	4000	0	FULL
Z04	Ward 5	4000	2000	14.00	4444	4000	0	FULL
Z01	Municipal Hospital	4000	3000	11.80	4082	4000	0	FULL
Z02	Central Flood Shelter	3500	2500	11.60	529	508	2992	BELOW MIN
Z05	School Emergency Shelter	2600	1600	11.50	0	0	2600	UNSERVED
Z06	Fire and Emergency Service	2000	1500	10.90	0	0	2000	UNSERVED

=====

Priority Order:

1. Z03 - Ward 3 (Priority 14.00)

2. Z04 - Ward 5 (Priority 14.00)

3. Z01 - Municipal Hospital (Priority 11.80)

4. Z02 - Central Flood Shelter (Priority 11.60)

5. Z05 - School Emergency Shelter (Priority 11.50)

6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary

=====

Total Requested Water : 20100.00 L

Total Allocated Water : 13500.00 L

Total Effective Water Delivered : 12508.30 L

Total Unresolved Requirement : 7591.70 L

Remaining Water : 0.00 L

=====

Service Conditions

=====

Full Requirement Satisfied : 3 zone(s)

Minimum Requirement Satisfied: 0 zone(s)

Below Minimum : 1 zone(s)

Completely Unserved : 2 zone(s)

=====

EMERGENCY POTABLE WATER ALLOCATION

=====

Lab_assignment.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fort

1#include <stdio.h>

2#include <string.h>

3#define TOTAL_ZONES 6

4struct Zone

5{

6char id[10];

7char name[40];

8char type[25];

9float requested;

10float minimum;

11float loss;

12int waiting;

13float priority;

14int order;

15float allocated;

16float lossAmount;

17float delivered;

18float shortage;

19char status[20];

20};

21/* Function Prototypes */

22void loadOriginalData(struct Zone

23void resetResults(struct Zone

Logs & others

Code::Blocks X Search results X Cccc X

File

Line

Message

=== Build file: "no

=== Build finished:

D:\CodeBlocks\Lab_assignment.c

D:\CodeBlocks\Lab_assignment.exe

Enter your choice: 5

===== CASE 5 =====

Residential zones are: Z03 and Z04

Enter residential Zone ID: Z03

Current distribution loss: 8.00%

Enter new distribution loss (%): 4

New loss of Z03 = 4.00%

=====

ALLOCATION RESULT

=====

ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z04	Ward 5	3800	1800	13.80	4318	3800	0	MINIMUM
Z03	Ward 3	4500	2000	13.60	4688	4500	0	FULL
Z01	Municipal Hospital	4000	3000	11.80	4082	4000	0	FULL
Z02	Central Flood Shelter	3500	2500	11.60	413	396	3104	BELOW MIN
Z05	School Emergency Shelter	2600	1600	11.50	0	0	2600	UNSERVED
Z06	Fire and Emergency Service	2000	1500	10.90	0	0	2000	UNSERVED

=====

Priority Order:

1. Z04 - Ward 5 (Priority 13.80)

2. Z03 - Ward 3 (Priority 13.60)

3. Z01 - Municipal Hospital (Priority 11.80)

4. Z02 - Central Flood Shelter (Priority 11.60)

5. Z05 - School Emergency Shelter (Priority 11.50)

6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary

=====

Total Requested Water : 20400.00 L

Total Allocated Water : 13500.00 L

Total Effective Water Delivered : 12696.18 L

Total Unresolved Requirement : 7703.82 L

Remaining Water : 0.00 L

=====

Service Conditions

=====

Full Requirement Satisfied : 2 zone(s)

Minimum Requirement Satisfied: 1 zone(s)

Below Minimum : 1 zone(s)

=====

Lab_assignment.c - Code::Blocks 25.03

File Edit View Search Project Build Debug For...

1#include <stdio.h>

2#include <string.h>

3#define TOTAL_ZONES 6

4struct Zone

5{

6char id[10];

7char name[40];

8char type[25];

9float requested;

10float minimum;

11float loss;

12int waiting;

13float priority;

14int order;

15float allocated;

16float lossAmount;

17float delivered;

18float shortage;

19char status[20];

20};

21/* Function Prototypes */

22void loadOriginalData(struct Zone

23void resetResults(struct Zone

Logs & others

Code::Blocks X Search results X Cccc X

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== Build file: "no

== Build finished:

D:\CodeBlocks\Lab_assignment.c

D:\CodeBlocks\Lab_assignment.exe

Enter your choice: 6

===== CASE 6 =====

Previously underserved zones from Case 1:

Z02 - Central Flood Shelter

Z05 - School Emergency Shelter

Z06 - Fire and Emergency Service

Enter Zone ID from the underserved list: Z02

Current waiting cycles: 1

Enter new waiting cycles: 4

Waiting cycles of Z02 changed to 4.

=====

ALLOCATION RESULT

=====

ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z02	Central Flood Shelter	3500	2500	17.60	3646	3500	0	FULL
Z04	Ward 5	3800	1800	13.80	4318	3800	0	MINIMUM
Z03	Ward 3	4500	2000	13.20	4891	4500	0	FULL
Z01	Municipal Hospital	4000	3000	11.80	645	632	3368	BELOW MIN
Z05	School Emergency Shelter	2600	1600	11.50	0	0	2600	UNSERVED
Z06	Fire and Emergency Service	2000	1500	10.90	0	0	2000	UNSERVED

=====

Priority Order:

1. Z02 - Central Flood Shelter (Priority 17.60)

2. Z04 - Ward 5 (Priority 13.80)

3. Z03 - Ward 3 (Priority 13.20)

4. Z01 - Municipal Hospital (Priority 11.80)

5. Z05 - School Emergency Shelter (Priority 11.50)

6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary

=====

Total Requested Water : 20400.00 L

Total Allocated Water : 13500.00 L

Total Effective Water Delivered : 12431.79 L

Total Unresolved Requirement : 7968.21 L

Remaining Water : 0.00 L

Service Conditions

=====

Lab_assignment.c - Code::Blocks 25.03

File Edit View Search Project Build Debug For...

1#include <stdio.h>

2#include <string.h>

3#define TOTAL_ZONES 6

4struct Zone

5{

6char id[10];

7char name[40];

8char type[25];

9float requested;

10float minimum;

11float loss;

12int waiting;

13float priority;

14int order;

15float allocated;

16float lossAmount;

17float delivered;

18float shortage;

19char status[20];

20};

21/* Function Prototypes */

22void loadOriginalData(struct Zone

23void resetResults(struct Zone

Logs & others

Code::Blocks X Search results X Cccc X

FileLineMessage

== Build file: "no

== Build finished:

D:\CodeBlocks\Lab_assignment.c

D:\CodeBlocks\Lab_assignment.exe

10. Exit

Enter your choice: 7

===== CASE 7 =====

Independent Test Case

Enter any available water quantity: 5000

=====

ALLOCATION RESULT

=====

ID	Zone	Request	Minimum	Priority	Allocated	Delivered	Shortage	Status
Z04	Ward 5	3800	1800	13.80	4318	3800	0	MINIMUM
Z03	Ward 3	4500	2000	13.20	682	627	3873	BELOW MIN
Z01	Municipal Hospital	4000	3000	11.80	0	0	4000	UNSERVED
Z02	Central Flood Shelter	3500	2500	11.60	0	0	3500	UNSERVED
Z05	School Emergency Shelter	2600	1600	11.50	0	0	2600	UNSERVED
Z06	Fire and Emergency Service	2000	1500	10.90	0	0	2000	UNSERVED

=====

Priority Order:

1. Z04 - Ward 5 (Priority 13.80)

2. Z03 - Ward 3 (Priority 13.20)

3. Z01 - Municipal Hospital (Priority 11.80)

4. Z02 - Central Flood Shelter (Priority 11.60)

5. Z05 - School Emergency Shelter (Priority 11.50)

6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary

=====

Total Requested Water : 20400.00 L

Total Allocated Water : 5000.00 L

Total Effective Water Delivered : 4427.27 L

Total Unresolved Requirement : 15972.73 L

Remaining Water : 0.00 L

Service Conditions

=====

Full Requirement Satisfied : 0 zone(s)

Minimum Requirement Satisfied: 1 zone(s)

Below Minimum : 1 zone(s)

Completely Unserved : 4 zone(s)

8. Limitations

- Fixed at 6 zones (TOTAL_ZONES); not dynamically extensible.
- Priority weights are a proposed heuristic, not a standard mathematical formula.
- Uses float data types, so minor floating-point rounding differences may appear.
- Single allocation pass — no second-round redistribution of leftover water.
- Assumes loss %, requests, and minimums are accurate and static during execution.
- Cases 5 and 6 allow only one zone change per run.
- Data resets every programme run (in-memory state, not persistently saved).

9. Conclusion

The programme models a realistic flood-response water allocation problem by combining urgency, waiting time, demand gap, and loss penalty into one priority score, then distributing water in three tiers (full → minimum → remaining). Testing across seven cases confirms consistent, logical behaviour, making the approach a practical and extensible solution for constrained-resource distribution.

10. References

- [1] E. Balagurusamy, Programming in ANSI C, McGraw-Hill Education. (confirm exact edition/year before submission)
- [2] OpenAI, ChatGPT (large language model), OpenAI. [Online]. Available: <https://chat.openai.com>